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#include<stdio.h>
#include<graphics.h>
#include<conio.h>
#include<math.h>

int round(double temp)
{
    if(temp>0)
        return int(temp+0.5);
    else
        return int(temp-0.5);
}

void DDA_Line(double x1,double y1,double x2,double y2 )
{
    double x=x1,y=y1;
    double dx=x2-x1;
    double dy=y2-y1;
    double steps=fabs(dx)>fabs(dy)?fabs(dx):fabs(dy);
    double xinc=dx/steps;
    double yinc=dy/steps;
    for(int i=0;i<steps;++i)
    {
        putpixel(round(x),round(y),RED);
        x+=xinc;
        y+=yinc;
    }
}

void Bresenham_Line(int x1, int y1, int x2, int y2)
{
    int k=(y2-y1)/(x2-x1);
    int x=x1,y=y1;
    int steps,t,xinc,yinc;
    if(k<=1)
    {
        steps=x2-x1;
        t=2*k-1;
        xinc=1;
        for(int i=0;i<steps;++i)
        {
            putpixel(x,y,RED);
            if(t>=0)
            {

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        t+=2*k-2;
        yinc=1;
    }
    else
    {
        t+=2*k;
        yinc=0;
    }
    x+=xinc;
    y+=yinc;
}
}
else
{
    steps=y2-y1;
    t=2/k-1;
    yinc=1;
    for(int i=0;i<steps;++i)
    {
        putpixel(x,y,RED);
        if(t>=0)
        {
            t+=2/k-2;
            xinc=1;
        }
        else
        {
            t+=2/k;
            xinc=0;
        }
        x+=xinc;
        y+=yinc;
    }
}
}

```

```

int main( )
{
    initgraph(800,600);
    setbkcolor(WHITE);
    cleardevice( );

    DDA_Line(20,20,500,500);
    Bresenham_Line(30,30,400,400);
}

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    getch( );  
    closegraph( );  
}
```